

TUMOR LOCALIZATION · CHIP-DESIGN STUDY

# Frequency Reduction

How narrow can the measurement band go before localization degrades?

CNN classification · raw S-parameters · LOSO · measured A3 phantom data

**1–5 GHz**

recommended band

**1/2**

the bandwidth of the full sweep

**100%**

F5 accuracy retained (full array)

# Smaller bandwidth, simpler chip

An integrated version of this system lives or dies on front-end complexity: a narrower sweep means simpler synthesizers, antennas and calibration. The question is what accuracy that costs, and where the band should sit.

- 1 WHERE does the information live?** slide a 2 GHz window across 0.1–8 GHz; full 4-antenna array
- 2 HOW WIDE must the band be?** expand 2 → 3 → 4 → 5 GHz around the winning region
- 3 Does it combine with reduced hardware?** winning band × fewer antennas / reflection-only
- 4 Is the band choice robust?** repeat the whole sweep with a single antenna (S11 only)

*Every cell: CNN classifier, raw mag+phase input, 100-epoch leave-one-session-out, per-position vote.*

## A 2 GHz window: the low-mid band wins

| Band (GHz)     | Width   | Empty       | A3 + F4     | A3 + F5     |
|----------------|---------|-------------|-------------|-------------|
| 0.1 – 2        | 2 GHz   | 99.4        | 96.2        | 90.9        |
| <b>2 – 4</b>   | 2 GHz   | <b>99.4</b> | <b>97.4</b> | <b>92.9</b> |
| 4 – 6          | 2 GHz   | 99.4        | 89.1        | 71.7        |
| 6 – 8          | 2 GHz   | 99.4        | 97.4        | 69.7        |
| 0.1 – 8 (full) | 7.9 GHz | 99.3        | 97.4        | 100.0       |

All 16 S-parameters. LOSO per-position vote accuracy (%). Highlighted row = best 2 GHz window (mean across phantoms).

The 2-4 GHz window wins. Above 4 GHz, F5 collapses to ~70%: the large lossy glandular insert attenuates high-frequency energy before it reaches interior positions. The empty phantom is band-agnostic.

## Expanding around 2-4 GHz: the knee is at 4 GHz

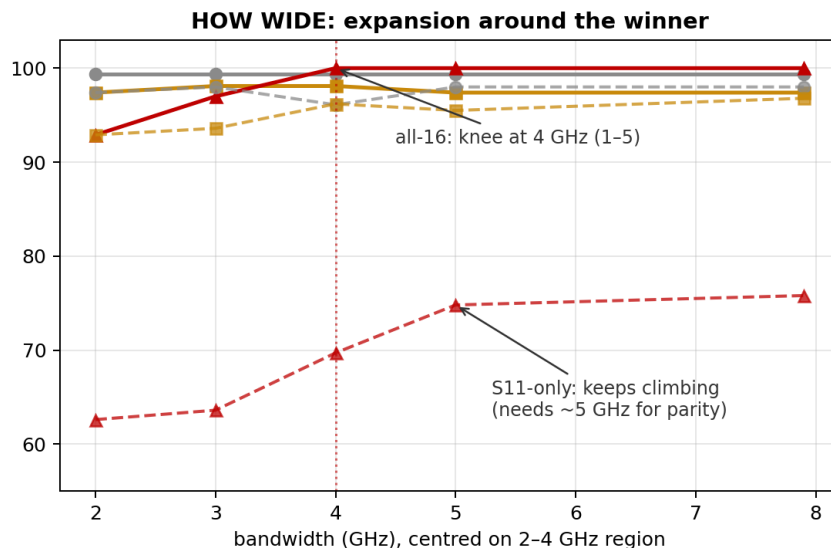
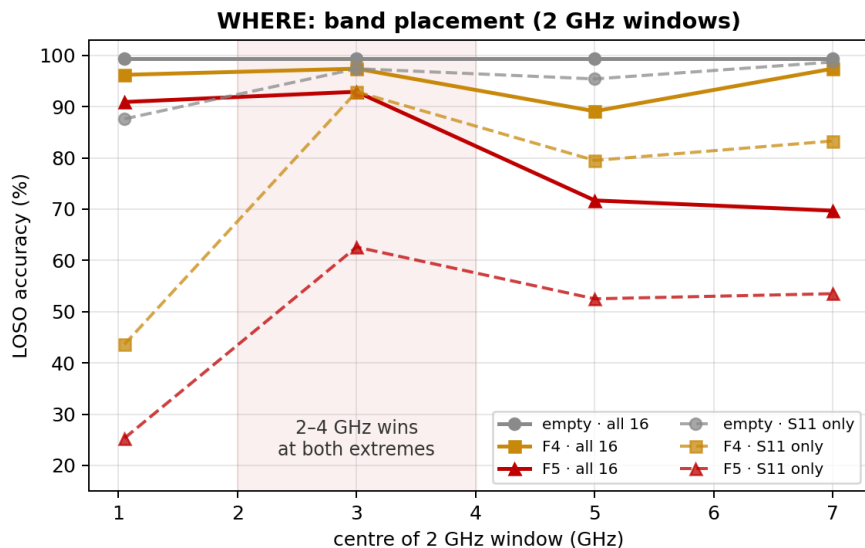
| Band (GHz)     | Width   | Empty       | A3 + F4     | A3 + F5      |
|----------------|---------|-------------|-------------|--------------|
| 2 – 4          | 2 GHz   | 99.4        | 97.4        | 92.9         |
| 1.5 – 4.5      | 3 GHz   | 99.4        | 98.1        | 97.0         |
| <b>1 – 5</b>   | 4 GHz   | <b>99.4</b> | <b>98.1</b> | <b>100.0</b> |
| 0.5 – 5.5      | 5 GHz   | 99.4        | 97.4        | 100.0        |
| 0.1 – 8 (full) | 7.9 GHz | 99.3        | 97.4        | 100.0        |

*All 16 S-parameters. Windows centred on the winning 2-4 GHz region.*

1-5 GHz (4 GHz bandwidth, 3 GHz centre) recovers full-band accuracy on every phantom, including 100% on F5. Going wider adds nothing; going narrower costs F5 3-7 points. Half the bandwidth, zero accuracy cost.

# Placement and width, at both hardware extremes

CNN classification vs frequency band: full array (solid) vs single antenna (dashed)



Solid = all 16 S-parameters · dashed = single antenna (S11 only). Left: where the 2 GHz window sits. Right: accuracy vs bandwidth around 2-4 GHz.

### STEP 3 · COMBINED REDUCTIONS

## The 1-5 GHz band with reduced hardware

| Hardware @ 1-5 GHz          | Empty | A3 + F4 | A3 + F5 | F5 @ full band |
|-----------------------------|-------|---------|---------|----------------|
| 4 antennas, reflection only | 100.0 | 97.4    | 94.9    | 100.0          |
| 2 antennas (1 & 3), full S  | 99.3  | 92.9    | 92.9    | 96.0           |
| 2 antennas, reflection only | 99.3  | 92.9    | 84.8    | 90.9           |
| 1 antenna (S11)             | 96.1  | 96.2    | 69.7    | 75.8           |

The reductions compound on F5: a band cut that is free with the full array costs 3-6 points once antennas or transmission paths are also removed. Best reduced operating point: 4 antennas, reflection-only, 1-5 GHz = 95% on the hardest phantom.

## STEP 4 • ROBUSTNESS CHECK

# The same sweep with one antenna confirms the band

| Band (GHz)     | Width   | Empty       | A3 + F4     | A3 + F5     |
|----------------|---------|-------------|-------------|-------------|
| 0.1 – 2        | 2 GHz   | 87.6        | 43.6        | 25.3        |
| <b>2 – 4</b>   | 2 GHz   | <b>97.4</b> | <b>92.9</b> | <b>62.6</b> |
| 4 – 6          | 2 GHz   | 95.4        | 79.5        | 52.5        |
| 6 – 8          | 2 GHz   | 98.7        | 83.3        | 53.5        |
| 0.5 – 5.5      | 5 GHz   | 98.0        | 95.5        | 74.8        |
| 0.1 – 8 (full) | 7.9 GHz | 98.0        | 96.8        | 75.8        |

*S11 only. Note 0.1-2 GHz: fine with the full array (90.9 on F5), catastrophic alone (25.3).*

2-4 GHz wins again, so the band choice is robust; but with one antenna there is no knee - accuracy climbs to ~5 GHz. Antennas and bandwidth are partially interchangeable information budgets.

# Chip band specification

## **Spec: 1-5 GHz**

4 GHz bandwidth, 3 GHz centre. Full-band accuracy on every phantom with the array; the anchor for any hardware configuration.

## **Fallback: 1.5-4.5 GHz**

3 GHz bandwidth costs only 1-3 points on the hardest phantom; 2-4 GHz is the floor (-7 on F5).

## **Avoid: above 5 GHz and below 1 GHz alone**

High bands carry no F5 information (insert attenuation).  
Pure low band collapses without array diversity.

## **Design principle**

Antennas and bandwidth trade against each other. Cut one aggressively, not both: 4-ant reflection-only @ 1-5 GHz keeps 95% on F5.

# New constraint, new question: where does it fail?

Follow-up task: stay entirely below 4 GHz (easier chip) and push the bandwidth down until the model breaks. That needs a definition of 'broken':

## DEGRADED

mean below 90% of that phantom's full-band accuracy

## UNSTABLE

fold-to-fold sigma of 10 points or more: works one session, fails the next

## BROKEN

mean below 50%: wrong more often than right (the headline break)

## DEEP

mean below 25%: approaching uselessness (still ~10x chance)

Method: descend bandwidth 3 -> 2 -> 1 -> 0.5 -> 0.25 -> 0.1 -> 0.05 GHz with the best placement at every width, plus a nine-window placement scan at 0.25 GHz. All below 4 GHz.

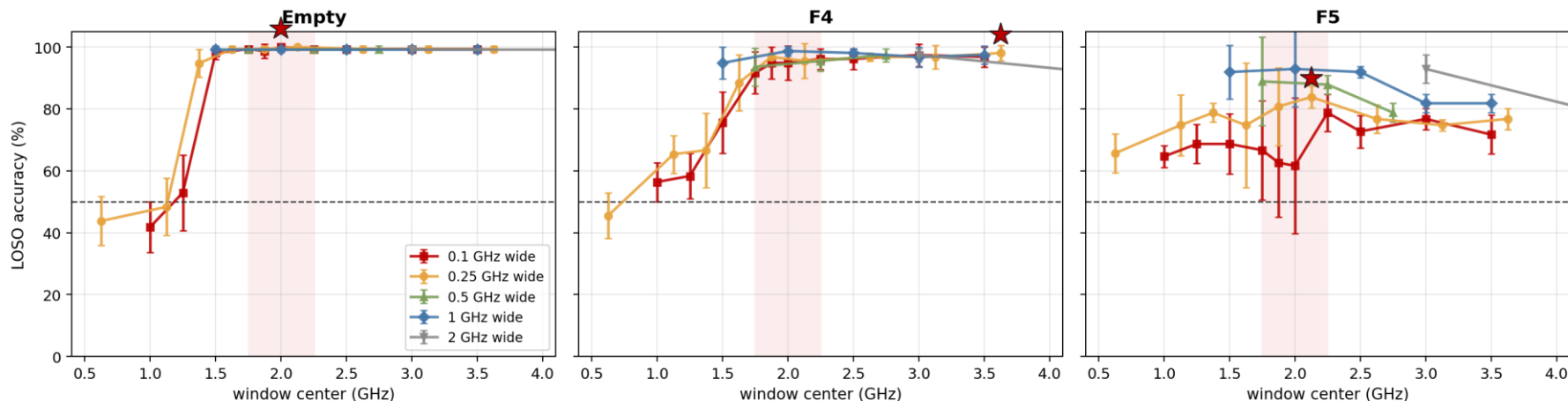
## A 0.25 GHz window: placement is everything

| Window (GHz) | Empty       | A3 + F4     | A3 + F5     |
|--------------|-------------|-------------|-------------|
| 0.5 - 0.75   | 43.8 ± 7.9  | 45.5 ± 7.4  | 65.7 ± 6.3  |
| 1 - 1.25     | 48.4 ± 9.3  | 65.4 ± 6.1  | 74.7 ± 9.7  |
| 1.25 - 1.5   | 94.8 ± 4.5  | 66.7 ± 12.0 | 78.8 ± 3.0  |
| 1.5 - 1.75   | 99.3 ± 1.1  | 88.5 ± 9.0  | 74.7 ± 20.2 |
| 1.75 - 2     | 99.3 ± 1.1  | 96.8 ± 1.3  | 80.8 ± 12.6 |
| 2 - 2.25     | 100.0 ± 0.0 | 95.5 ± 5.7  | 83.8 ± 3.5  |
| 2.5 - 2.75   | 99.3 ± 1.1  | 96.8 ± 1.3  | 76.8 ± 4.6  |
| 3 - 3.25     | 99.3 ± 1.1  | 96.8 ± 3.8  | 74.7 ± 1.7  |
| 3.5 - 3.75   | 99.3 ± 1.1  | 98.1 ± 2.5  | 76.8 ± 3.5  |

Best slot: 2-2.25 GHz (highlighted; also far more stable than 1.75-2). Red rows: below ~1.25 GHz even the EMPTY phantom breaks the 50% line. Bandwidth is survivable; bad placement is not.

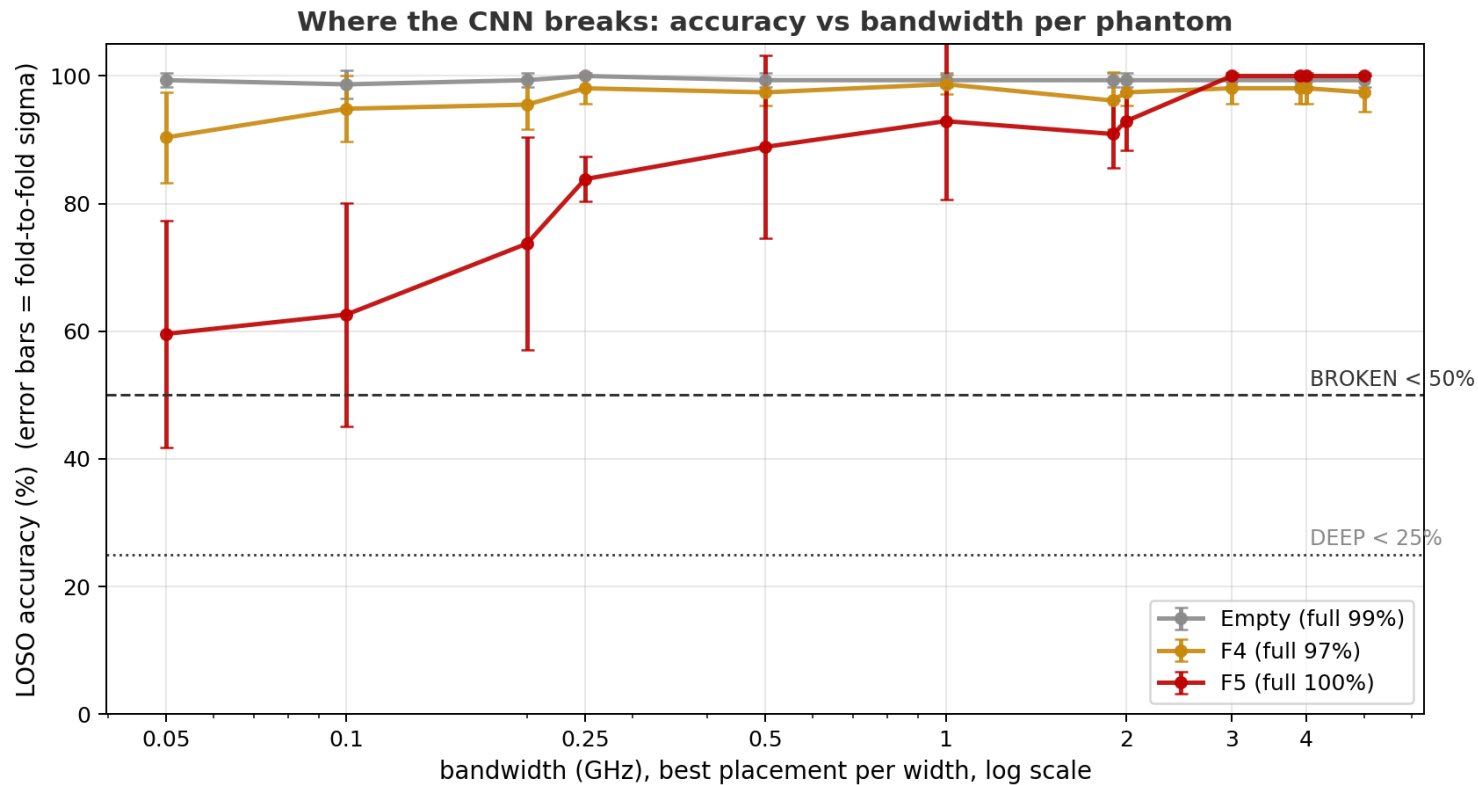
# Each phantom has its own best frequency

Where the information lives: train on ONLY one window, slide it across the spectrum. Star = best narrow slot per phantom (each is different)



Sufficiency scan: train on ONLY one window and slide it across the spectrum. Best narrow slot differs by phantom: empty ~2.0 GHz (indifferent above 1.45), F4 ~3.0 (small target rewards higher frequency), F5 ~2.2-2.3 (large lossy insert forces low). The descent's 1.85-1.9 GHz slot was NOT F5's best (62.6 there vs 78.8 at 2.2-2.3), so true narrow-band floors are even higher. Pick the chip band for the worst-case target.

# Best placement at every width, down to 0.05 GHz



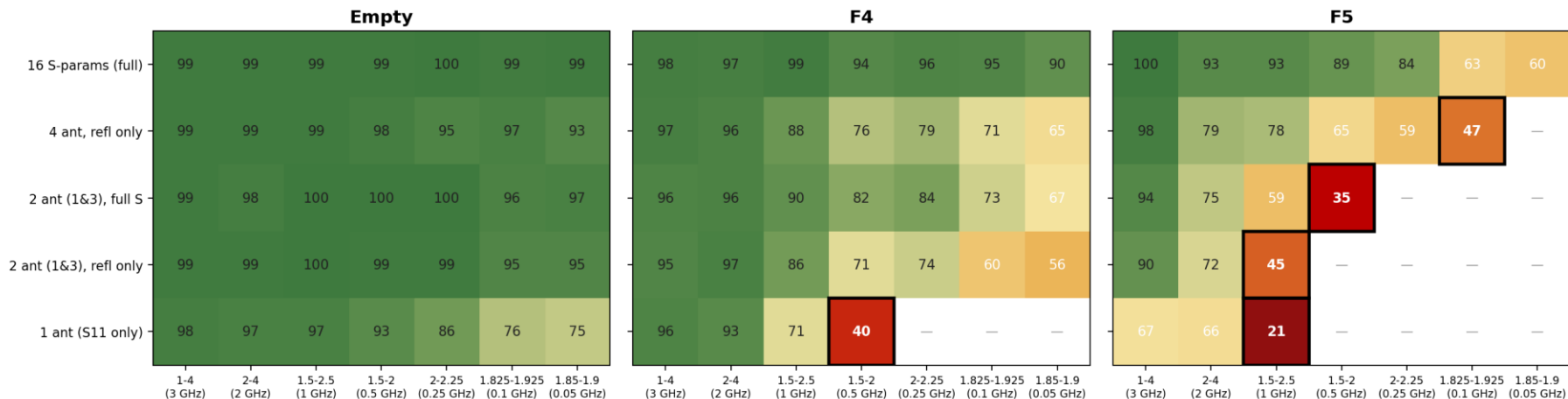
# Where each phantom breaks

| Failure tier                 | Empty      | A3 + F4    | A3 + F5     |
|------------------------------|------------|------------|-------------|
| Degraded (<90% of full band) | never      | never      | 0.5 GHz     |
| Unstable (fold sigma >= 10)  | never      | never      | 1 GHz       |
| Broken (<50%)                | never      | never      | never       |
| Floor at 0.05 GHz (50 MHz)   | 99.3 ± 1.1 | 90.4 ± 7.1 | 59.6 ± 17.8 |

With all 16 S-parameters and a well-placed window, bandwidth alone cannot push any phantom below 50%: even a 50 MHz sliver holds 99 / 90 / 60. F5 fails softly (degraded, then session-unstable below 1 GHz). Breaking for real requires combining the band cut with hardware reduction; that adaptive descent is Part 3.

# The full break map: 5 hardware levels x 7 bands

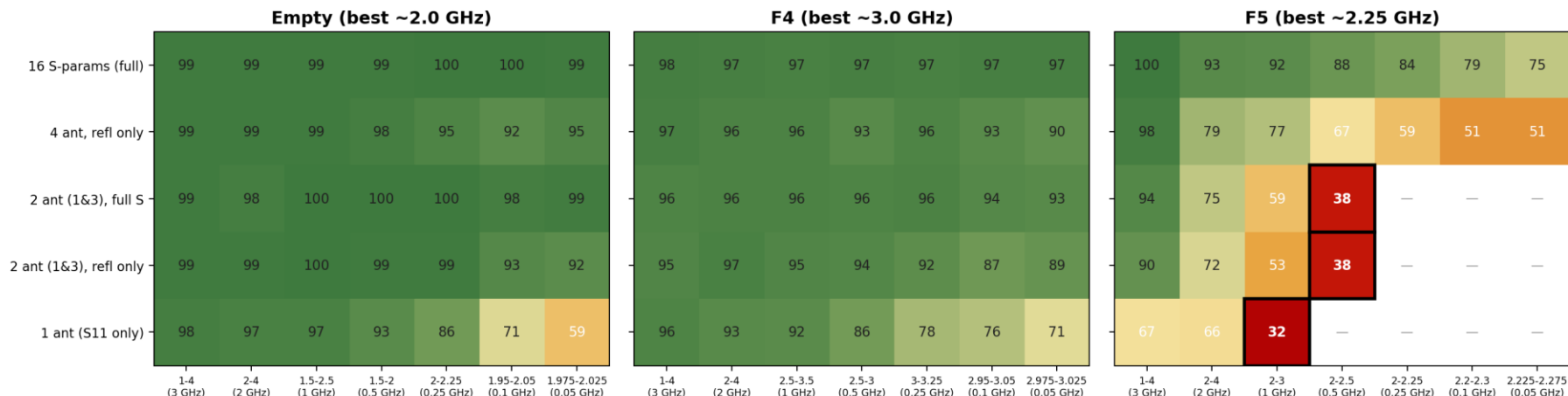
LOSO accuracy (%) across hardware reduction x band narrowing — black box = BROKEN (<50%), — = skipped after break



Adaptive descent: bands narrow left to right at the best placement per width; once a phantom scores below 50% (black box) it stops descending at that hardware level. Blank cells were skipped after the break. Empty stays green everywhere.

# Same descent, but each phantom aims at its own best frequency

Break map v2 - each phantom's bands converge on ITS OWN best frequency. Black box = BROKEN (<50%), — = skipped after break



Re-centered ladders (empty -> 2.0 GHz, F4 -> 3.0, F5 -> 2.25). F4's single-antenna break vanishes entirely (70% at 50 MHz vs broken at 0.5 GHz before): its 'break' was bad placement, not missing information. F5's floors rise ~10-15 points and it survives one step further, but its reduced-hardware breaks remain - those are physics. Caveat: best centers came from full-array scans and shift slightly at reduced hardware (empty on S11 alone did better at the old 1.85-1.9 slot).

## Where each phantom finally breaks (<50%)

| Hardware                            | Empty            | A3 + F4       | A3 + F5          |
|-------------------------------------|------------------|---------------|------------------|
| 16 S-params (full array)            | never            | never         | never (floor 60) |
| 4 antennas, reflection only         | never            | never         | 0.1 GHz (47%)    |
| 2 antennas (1 & 3), full S          | never            | never         | 0.5 GHz (35%)    |
| 2 antennas (1 & 3), reflection only | never            | never         | 1 GHz (45%)      |
| 1 antenna (S11 only)                | never (floor 75) | 0.5 GHz (40%) | 1 GHz (21%)      |

- Difficulty ladder is perfectly ordered: empty never breaks (75% even on S11 + 50 MHz); F4 breaks only at the most extreme cut; F5 breaks at every reduced-hardware level.
- Antennas and bandwidth trade against each other: each hardware cut moves F5's breaking point to a wider band (0.1 -> 0.5 -> 1 GHz).
- Chip takeaway: with 2 antennas keep at least ~2 GHz of band (in the 1-4 GHz region); with the full array, bandwidth is nearly free.